

$$\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ \textcircled{2} & 3 & 4 & 5 \\ 5 & 6 & 7 & 8 \end{array}$$

Fórmula

Linha para zerar = $\frac{\text{Num para zerar}}{\text{Num que tá em cima}} \times \text{Linha que tá em cima}$

$$L_2 = L_2 - \frac{2}{1} \times L_1$$

$$2 - 2 \times 1 = 0$$

$$3 - 2 \times 2 = 3 - 4 = -1$$

$$4 - 2 \times 3 = 4 - 6 = -2$$

$$5 - 2 \times 5 = 5 - 10 = -5$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 0 & -1 & -2 & 1 \\ \textcircled{5} & 6 & 7 & 8 \end{array} \rightarrow$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 0 & -1 & -2 & 1 \\ 0 & \textcircled{-4} & -8 & -2 \end{array} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -2 & 1 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$-1x + 2y + 3z = 2$$

$$-1y - 2z = 1$$

$$= 6$$

Impossible

$$L_3 = L_3 - \frac{5}{1} \times L_1$$

$$5 - 5 \times 1 = 0$$

$$6 - 5 \times 2 = 6 - 10 = -4$$

$$7 - 5 \times 3 = 7 - 15 = -8$$

$$8 - 5 \times 2 = 8 - 10 = -2$$

$$L_3 = L_3 - 4 \times L_2$$

$$\begin{aligned} -4 \times (-1) &= 4 \\ -4 + 4 &= 0 \end{aligned}$$

$$-4 - 4 \times (-1) = 0 \quad \frac{-4}{-1} = \frac{4}{1} = 4$$

$$-8 - 4 \times (-2) = -8 + 8 = 0$$

$$-2 - 4 \times 1 = -2 - 4 = -6$$